

IAC-25-B4

HORUS: A High-Resolution Earth Observation Mission in Very Low Earth Orbit

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Abstract

Interest in Very Low Earth Orbit (VLEO) is significantly increasing in Earth Observation missions due to its unique advantages. Operating at this low altitude enables cost-effective high-resolution imaging, improved revisit time and a cleaner orbital environment with reduced space debris. However, only few missions have flown in VLEO because of its challenging environmental conditions. Indeed, the high atmospheric drag and the concentration of atomic oxygen pose significant problems, strongly influencing the mission lifetime and requiring for innovative design solutions.

The High-resolution Orbiting Reconnaissance and Universal Surveillance (HORUS) is a mission concept developed over a four-month period at Politecnico di Milano. Designed as an advanced Earth Observation mission operating in VLEO, HORUS aims to deliver ultra-high-resolution images with minimal latency. The mission is engineered to achieve high performances, featuring a spatial resolution below one meter, a revisit time of less than one day, and the ability to downlink images within one hour from acquisition. HORUS is entirely based on European technology, from the spacecraft components to the ground segment and launch systems. Moreover, since only a few missions have already flown in VLEO, it will play a crucial role in validating satellite design, materials, and technologies for this environment. However, the true strength of HORUS lies in its flexibility and scalability, as it can cover a wide range of target areas based on customer requirements, ensuring operational readiness within just two months from the request. The baseline architecture of HORUS consists of a single space drone, which is sufficient when the required coverage area is relatively small. However, for larger coverage needs, multiple drones can be deployed in formation flights, allowing the mission to be efficiently scaled according to operational requirements.

Designed to be launched in 2029, HORUS versatility makes it well suited for a wide range of applications, including environmental monitoring, real-time disaster response, and dual-use operations. The design study conducted demonstrates the feasibility of developing a compact, cost-effective space drone with a mass of approximately 100 kg, capable of achieving the ambitious mission objectives, tackling VLEO criticalities and exploiting its advantages.

This paper discusses the HORUS architecture, highlighting design solutions that enable the achievement of its mission goals. It also provides a design overview of all the subsystems, with their key challenges, critical aspects, and innovative solutions. The final purpose is to present HORUS mission concept, showcasing the potential of Earth Observation small satellites operating at those very low altitudes.

1 Introduction

Very Low Earth Orbit (VLEO) is gaining attention in the Earth observation mission scenario due to the significant advantages it offers. However, the extreme environmental conditions in VLEO have historically made this orbital altitude unattractive for mission operations. Indeed, fully leveraging the benefits of VLEO requires a mission design approach that deviates substantially from conventional Low Earth Orbit (LEO) practices, addressing unique challenges.

The purpose of this paper is to present a comprehensive mission feasibility study for High-resolution Orbiting Reconnaissance and Universal Surveillance (HORUS), an innovative mission specifically designed to operate effectively in VLEO. It exemplifies how meticulous mission design and careful planning can transform even a hostile environment into an attractive operational domain.

HORUS is an Earth observation satellite capable of delivering ultra-high-resolution imagery with minimal latency between image acquisition and data downlink. Through detailed mission analysis and optimized launch planning, it can achieve high flexibility, quickly responding to any customer request regarding targeted observation areas. Given the innovative nature of the mission, extensive consideration has been given on integrating advanced technologies, both in launch vehicles and onboard hardware. Furthermore, HORUS will serve as a platform to validate the Technology Readiness Level (TRL) of these technologies, thereby establishing a foundation for future missions operating at similar low altitudes.

This paper is structured as follows. **Section 2** reviews the state of the art for previous missions conducted in VLEO. **Section 3** provides a detailed overview of HORUS mission objectives and high-level goals. **Section 4** examines the harsh VLEO environment, emphasizing the main challenges it presents and highlighting key design choices. **Sections 5 and 6** delve deeper into the HORUS mission design, beginning with a general overview of the architecture and mission budgets, followed by an analysis of each subsystem, detailing the specific design choices, challenges encountered, and innovative solutions adopted. **Section 7** focuses on mission cost analysis, covering all phases from Phase 0 to Phase E, and underscores the feasibility of maintaining sustainable costs despite mission complexity. **Sections 8 and 9** conclude the paper with the exploration of potential HORUS applications and future developments.

2 State of the art

Although VLEO has attracted increasing interest in recent years, its state of the art remains limited due to the harsh environmental conditions.

To address these challenges, most missions have medium to low mass budgets, in order to favor compact and lightweight designs that can better cope with VLEO environment. Indeed, the design of small satellites is prevalent, as demonstrated by missions like Colony-1 [1], SuperDove [2], SkySat [3], and QARMAN [4]. However, there is an example of a satellite that goes against all these design paradigms: European Space Agency (ESA)'s GOCE [5] [6]. It is a satellite with a mass exceeding 1000 *kg* operating in VLEO, justified by its scientific objectives and payload requirements. Finally, many mission concepts have been studied for operation in this environment, in order to assess feasibility, mission needs, and overall practicality. Some examples include THOR [7] and DMC-HD [8].

One of the key challenges for VLEO missions is the propulsion, as it is required to continuously counteract aerodynamic drag. Electric propulsion is commonly employed due to its suitability for continuous thrust, with typical lifetimes of 4 to 5 years. In contrast, missions without propulsion, like Colony-1, decay within a few months [1]. Recent technological developments include Air-Breathing Electric Propulsion (ABEP), which utilizes residual atmospheric gases as propellant, [9] and High Performance Green Propulsion (HPGP) [10], offering improved efficiency and reduced toxicity.

The most valuable instruments for VLEO missions are those focused on Earth Observation and atmospheric analysis. The low altitude enables detailed studies of the Earth's gravity field and upper atmosphere [5] [6]. Additionally, imaging systems benefit from the reduced orbital height, achieving significantly higher spatial resolution. However, satellites operating in these regions face challenges due to their high ground speed, which makes data downlink difficult. A promising solution to this issue is the use of relay satellites.

Attitude Determination and Control Subsystem (ADCS) in these regions typically employs a combination of reaction wheels and magnetorquers [5]. For satellites operating below 300 km, the presence of aerodynamic surfaces can also assist in passive stabilization thanks to the atmospheric drag [1] [8].

Power generation relies on high-efficiency solar cells, typically mounted on the satellite body [5]. If struc-

tural appendices are available, they are also used to host additional solar cells [5]. Due to the high frequency of Sun-eclipse cycles, onboard batteries are frequently employed to ensure continuous power supply.

3 Mission Objectives

To accomplish the ambitious goals of the HORUS mission, several key mission objectives have been defined. This section details these objectives, emphasizing the corresponding high-level goals and highlighting the primary subsystems most impacted by these requirements.

To provide specific area coverage of the Earth with high resolution, low latency, and minimum time resolution.

HORUS aims to deliver Earth imagery with a **spatial resolution below one meter**. This high-resolution requirement significantly impacts the payload design and places stringent demands on the Attitude and Orbit Control System (AOCS), which must ensure precise pointing stability during imaging operations. Additionally, to ensure optimal temporal coverage, HORUS is designed to revisit target areas at least once per day. Frequent revisits facilitate timely monitoring and rapid response, particularly crucial for applications such as disaster management and dual-use operations.

To provide fast data download.

Another critical objective is achieving low-latency data transmission, specifically the capability to **downlink imagery within one hour from acquisition**. Meeting this goal requires precise antenna pointing, directly impacting the AOCS design, and the implementation of a robust Telemetry, Tracking, and Command (TTMTC) subsystem. The On-Board Data Handling (OBDH) system must efficiently manage and store large volumes of high-resolution data prior to transmission. This near real-time transmission capability is essential for timely risk mitigation and prompt response.

To ensure flexibility in terms of target covered.

Flexibility is a key feature of the HORUS mission. Given the different types of potential applications, the satellite must be capable of rapidly adapting coverage areas based on evolving stakeholder needs. Furthermore, HORUS is expected to remain operational for as long as necessary or until the conclusion of specific observation events. Achieving this operational flexibility in the harsh VLEO environment, characterized by significant atmospheric drag and atomic oxygen exposure, is extremely challenging, posing stringent requirements

on the mission analysis, the propulsion and the Guidance, Navigation, and Control (GNC) subsystems.

To limit the time to launch and launch costs.

It is fundamental that HORUS can be launched quickly in response to emerging needs. Minimizing the launch time ensures coverage of critical areas during relevant windows. Additionally, reducing launch-related costs, especially in scenarios involving multiple satellite deployments, is a priority. HORUS is designed to be fully operational and **ready for launch within 1–2 months from initial client engagement**. This ambitious timeline requires meticulous launch planning and scheduling, as well as flexibility in choosing launch vehicles, aspects that will be explored further in Section 5.4.

To demonstrate the TRL for VLEO.

Operating in the challenging and relatively unexplored VLEO region presents significant risks. One key objective for HORUS is to demonstrate and validate technologies specifically suited to withstand these harsh conditions over extended durations. This approach provides a unique opportunity to validate innovative materials and subsystems, establishing benchmarks for future missions operating in VLEO environment. Additionally, HORUS will rely entirely on European-developed technologies. This includes the satellite platform, payloads, ground infrastructure, and launch vehicles.

4 Environment

Despite the several advantages offered by VLEO, its environment poses a series of challenges to be properly tackled and taken into consideration by the mission design. For instance, the high atmospheric density in VLEO significantly reduces satellite operational lifetimes and increases the risk of material degradation, mainly due to the concentration of atomic oxygen (ATOX), which accelerates corrosion. Furthermore, the design of the mission takes into account the disturbances of the geomagnetic field, which depend on the solar activity, and the effects of the different sources of thermal radiation.

4.1 ATOX degradation

When operating in VLEO, a spacecraft is exposed to the conditions of the lower thermosphere. Among these, one of the most critical is the presence of ATOX,

which can cause surface erosion, material degradation through ionization, as well as surface charging. At these altitudes, ATOX can constitute 30% to 60% of the atmosphere, depending on the orbital height. Additionally, atomic oxygen can penetrate cavities or gaps in satellites. This can create "hot spots" where ATOX concentration is high, increasing the risk of internal damage. It is crucial to predict these fluxes using analytical models and empirical data to ensure the survival of internal components and improve the overall satellite design. This is done mostly by using ATOX resistant and hardened surface materials, by positioning degradable materials only inside the main body of a spacecraft and by granting electrical continuity. In Table 1 can be found some typical ATOX material degradation values in high solar activity, worst case condition.

Material	Degradation [$mm/year$]
Kapton	0.995
Black Paint	0.123
SiO ₂ (coverglass)	0.0003
Aluminum	Negligible

Table 1: Worst case orbital ATOX degradation values [11][12]

4.2 High atmospheric density

At VLEO altitudes, atmospheric density is significantly higher with respect to most Earth-orbiting missions, varying drastically depending on the specific range of operation. This can have great effects which must be properly considered during both mission planning and spacecraft design. First off, it must be noted that the effects given by aerodynamic heating by the Earth's atmosphere are typically negligible at altitudes above 150 km, due to spacecraft sizes being significantly lower with respect to the mean free path of the thermosphere particles, which stands at around 38 m. Generally speaking, the increased density causes relatively rapid orbit decay and higher aerodynamic torque, which translate into an increase in Station Keeping (SK) fuel consumption. To get an idea of the decay encountered in VLEO, Fig. 1 shows the changing altitudes in time for a spacecraft on a circular orbit with a mass of 100 kg, dimensions of (30 x 30 x 100) cm, a cross section area of 0.09 m² and a ballistic coefficient of 2. To guarantee the safety of the mission, precautions must be taken to avoid any drastic issues. For example, placing sensitive components and electronics in protected areas of the configuration instead of directly

exposing them to the environment would be beneficial, as well as minimising the cross-section and maximizing the ballistic coefficient which is further discussed in Section 6.4.7

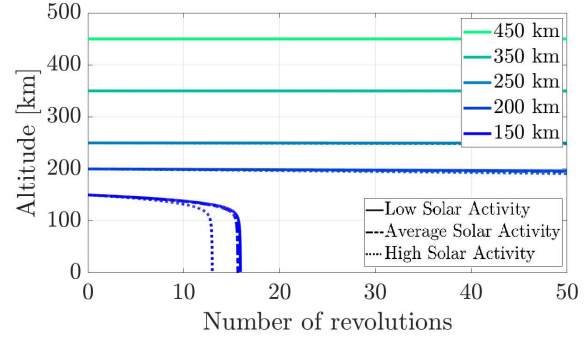


Figure 1: Orbital decay for different starting altitudes and different solar activities[13]

4.3 UV radiation

Even though Ultraviolet (UV) radiation levels are lower in VLEO due to the shielding given by the residual layer of atmosphere, their presence must still be evaluated given the critical mission environment. UV radiation is especially critical for the degradation it provokes in surface optical properties, especially absorptivity values. Given the lack of data available for VLEO altitudes, degradation analyses should be done in worst-case scenarios, considering no attenuation given by the atmosphere so as to be conservative in the design. In Table 2 can be seen some absorptivity values after 6 months in orbit. For later analyses, linear variations were considered for absorptivity degradation levels. UV resistant surface finishes and materials need to be considered for the spacecraft design.

Material	α , initial	α , 6 months
Silver Teflon	0.075	0.09
Opt. Solar Reflector	0.09	0.19
Black Kapton	0.59	0.68

Table 2: Absorptivity degradation values [14][15][16]

4.4 Charged particles

Within the space environment, trapped charged particles can have a negative impact on the hardware present on board of the spacecraft. These can provoke an increased possibility of Single Event Effects (SEEs) occurring, as well as major radiation damage for components. In Table 3, values associated to the radiation

levels encountered by components in VLEO can be appreciated as well as their design limits. As can be seen, for the listed parts and most hardware charged particles should not be an issue during the mission. For critical components, however, these factors can be further mitigated by positioning them inside the configuration and by implementing physical shields.

	Max. Rad.	Rad. after 1 Yr
Sun Sensor	30 <i>Krad</i>	1.343 <i>Krad</i>
PCDU	30 <i>Krad</i>	0.706 <i>Krad</i>

Table 3: Radiation levels encountered by components in VLEO in 1 year and maximum allowed component radiation levels [17]

4.5 Other factors

Due to the low orbital altitude, frequent eclipses need to be taken into account. The variable heat fluxes given by the environment require flexibility in the Thermal Control Subsystem (TCS) design and adaptable management of power needs.

Throughout the mission, the spacecraft encounters a varying magnetic field, which depends on the solar activity and the altitude, latitude and longitude of the spacecraft. The effects of the magnetic field must be taken into account when selecting the instruments to be used for attitude control purposes and when determining the configuration of the system electronics.

5 Architecture

The architecture solution represents HORUS baseline, serving as the framework upon which all mission elements are built. It integrates insights derived from the detailed subsystem analyses and aligns them with the constraints and requirements identified from mission objectives and high level goals.

5.1 Architecture choice

Upon the final choice of the architecture, a trade-off analysis with quantitative and qualitative parameters was conducted among three possible alternatives:

- **Scalable Architecture** : This approach consists of a single space drone, whose architecture can be scaled based on mission requirements. If the area to be covered is relatively small, a single drone may be sufficient. However, for larger areas, multiple launches of space drones, either

in flight formation or as part of a constellation, would be required.

- **Single Satellite Deployer**: In this architecture, a deployer is used to release drones directly into space as needed. This significantly reduces the time to fly but comes with both higher development costs and higher single point failure risk for the deployer.
- **Space Drones Constellation**: This solution involves maintaining a permanent constellation of drones in VLEO. The primary advantage is the ability to cover a vast area while significantly reducing the time to fly. However, a major challenge is the high maintenance cost associated with operating at such a low altitudes.

The three alternatives were then compared based on quantitative and qualitative parameters such as time to fly, total ΔV , ground visibility, cost, redundancy, revisit time, coverage, and TRL. Among the three, the one with the highest score was the scalable architecture, mainly because of its flexibility, lower costs, and redundancy features. Therefore, the final choice was to focus on designing a single space drone, and then to relate the number of drones to be employed to the required coverage and revisit time, also exploiting formation flights if needed. This choice enhances mission flexibility and scalability, according to customer needs. Further specifications on mission architecture in terms of launch strategy, coverage, and number of drones required are provided in Section 5.4 and Section 6.2. The satellite architecture concept is reported in Fig. 2, while, more insight on the overall system design rationale are presented in Section 6.

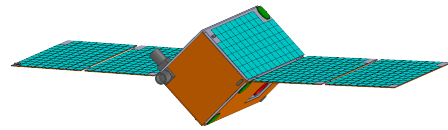


Figure 2: HORUS external configuration

5.2 Phases and modes

The lifespan of the HORUS mission shall be classified into distinct phases according to the functionalities and tasks the system has to perform. In the **Launch and Early Operations (LEOP)**, which is the first phase of the mission, the launch and the release from the launcher's deployer are performed. Following separa-

tion, the spacecraft powers up, activates its primary sensors and actuators to perform initial detumbling, and subsequently establishes its first communication link with the ground station. In **Commissioning Phase (COP)** the activation of all secondary sensors and actuators is performed, along with functional tests to ensure spacecraft health and operability. Following this phase, the mission proceeds to the **Observation Phase**. Once the satellite reaches its designated target orbit, it begins to perform the nominal functionalities of the mission which include: orbital control and station-keeping, Payload (PL) data acquisition, data processing and storage, communication with the GS and data transfer.

Upon reaching the end of its intended operational lifetime or upon completing its mission objectives, the spacecraft transitions into the **Disposal phase**. During this final phase, all the subsystems are progressively switched off, and the spacecraft is passivated and deactivated according to the ESA space debris mitigation requirements.

HORUS system works on the basis of 6 fundamental modes:

- **Activation Mode:** this mode activates the satellite after its deployment. During activation, the On-Board Computer (OBC) is powered on, and health checks are performed to verify system integrity.
- **Stand-by Mode:** it represents a go-through mode where the satellite fulfils only the Sun pointing requirement and charges the batteries.
- **Safe Mode:** mode in which the satellite enters if autonomously detects a predefined operating situation considered as off-nominal.
- **Orbit Control Mode:** this mode utilizes the thruster to perform various orbital manoeuvres, such as orbital insertion and station-keeping.
- **Imaging Mode:** this is the primary mission mode in which the payload is fully operational.
- **Telecom Mode:** in this mode, the satellite communicates with the ground station.

5.3 Budgets

In the context of space mission design, mass and power budgets are essential tools to analyze and evaluate the physical characteristics and requirements of each subsystem. They provide a comprehensive overview of the

spacecraft and help ensure consistency across the entire mission architecture.

5.3.1 Mass budget

The mass budget is defined according to ECSS standards [18], which provide guidelines for system engineering and mass margins. These margins are already included in Table 4.

Subsystem	Mass [kg]	Mass percentage
PL	12.600	15.64%
AOCS	11.077	13.75%
EPS	10.492	13.02%
OBDH	1.250	1.55%
PS	10.111	12.55%
Structure	16.766	20.81%
TCS	17.343	21.53%
TTMTC	0.930	1.15%
Total	80.564	100%

Table 4: Mass budget

According to ECSS standards [18], an additional 20% system-level margin is applied to the total mass, resulting in a final estimate of 96.676 kg. The Xenon propellant (~8 kg) is excluded from the budget, as ECSS standards refer to dry mass only. The Thermal Control Subsystem (TCS) is the heaviest subsystem, due to the thermal challenges of the selected low-altitude orbit. In contrast, the structural mass falls within typical values for LEO/VLEO missions [19][20].

5.3.2 Power budget

The power budget analysis is still conducted in accordance with the ECSS guidelines [18], supporting the correct sizing of both power generation and storage systems by detailing the power consumption of each unit across all operational modes. Firstly, the power budget is studied considering two main factors: the peak power, and the average power. Peak power refers to the maximum power required by a component when operating at full capacity, while average power represents the mean power consumption over an entire operational cycle. Clearly, it can be stated that the most power expensive mode for the HORUS mission is the Orbit Control Mode, with a **peak power of 895.56 W** and an **average power of 281.94 W**. These values reflect the architectural choices made during system design. In

this mode, the most expensive subsystem is the Propulsion Subsystem, since an electric thruster was chosen. Moreover, the ADCS is one of the most power-intensive subsystems in this mode, since the full sensors and actuators suite is required to meet the stringent pointing accuracy required during thrusting operations.

5.4 Launcher selection and strategy

The launcher selection is of utmost importance for HORUS mission. Indeed, time to fly and European technology requirements shrink the number of alternatives. At the beginning of 2025, the two European launchers available were the Ariane 6, with its two configuration, and the Vega-C. Both solutions allow access to almost any orbit inclination, but the former has lower costs and higher frequency, while the latter has higher flexibility but higher costs. Moreover, since HORUS is a lightweight satellite, for the Ariane 6 solution it might not have enough say in deciding the insertion orbit parameters, especially as part of a rideshare mission. Nevertheless, after evaluating all options against the mission's constraints and requirements, both launchers remain under consideration as potential solutions. Indeed, to date those are the only available options and both are required to fulfil the very tight time to fly requirement.

In the next years other European launchers may be available, increasing mission flexibility, reducing costs, and enhancing performances. Suitable solutions considered are the ones expected to launch before 2026 and allowing to store multiple satellites at once. Some of the most relevant options are RFA-One from Rocket Factory Augsburg (RFA), the Spectrum from Isar, and the Vega-E from Avio. However, for all listed launchers, the transition from the first commercial launch to the declared full capacity frequency might last several years and be subjected to changes. In Table 5 a comparison between the current available launchers and the next generation is reported.

Launcher	Debut Year	Frequency	Costs
Vega-C	In use	$\approx 5/\text{year}$	11.5 k€/kg
Ariane 62	In use	$\approx 4/\text{year}$	7.3 k€/kg
Ariane 64	In use	$\approx 4/\text{year}$	5.3 k€/kg
RFA-One	2025	$\approx 20/\text{year}$	2.3 k€/kg
Spectrum	2025	$\approx 12/\text{year}$	10-15 k€/kg
Vega-E	2026	$\approx 4/\text{year}$	11.5 k€/kg

Table 5: Launchers comparison [21], [22], [23]
Eventually, due to satellite very low altitude, in the

near future also airborne launchers could be employed. This solution entails much higher flexibility, with less time to flight and higher launch frequency, together with an increased decision-making authority over launch parameters. On the other hand this is still an under-development technology, therefore the actual performances, costs, and feature could not yet be reliably assessed.

Regarding the launch strategy, in general the philosophy for mission requiring more than one drone, is to initially launch a small batch, ensuring partial coverage and a rapid response time, and to complete the overall fleet, through successive launches. By doing so, the drones can be interchanged in a phased manner, without having to cycle a large number of drones at once.

6 Design

6.1 Payload

The Payload selection for the HORUS mission is crucial due to the high dependence on its performance. One of the most important objectives of the mission is to have a ground resolution below 1 *m*, which is evaluated in terms of GSD:

$$GSD = \frac{ph}{f} \quad (1)$$

where *p* is the photosite pitch, *h* is the altitude, and *f* is the focal length of the camera.

Moreover, the study must also consider the Swath Width (SW), which depends on altitude and scan angle.

The HORUS mission is built around the chosen payload, but still there are constraints on it, such as the operational range of temperature, the storage capacity, and the dimensions, crucial to assure a compact satellite. Eventually, a careful study must be conducted also on the spectrum bands of interest. Since the mission will have frequent Sun-eclipse cycles, the camera must guarantee also the possibility to take pictures with low illumination conditions. Therefore, a payload that can work in both the Visible Band and Infrared Band is preferred.

After having carefully considered all these characteristics and needs for the HORUS mission, the iSIM-90 dual channel [24] is selected. This choice follows a trade-off analysis between performances and dimension, ensuring the one meter resolution with good weight and great compactness.

Focal Length [mm]	1667
Max Alt. [km] ($GSD \leq 1\text{ m}$)	303
Mass [kg]	< 6
Dimensions [mm]	$308 \times 216 \times 115$
Power [W]	30.5
FoV [deg]	1.49 to 2.98
Swath [km] (@ Max Alt.)	7.88 to 15.76
Bands	PAN, VNIR (< 8)
Storage [GB]	500
Thermal Control	Active

Table 6: iSIM-90 (dual channel) technical characteristics.

Furthermore, a configuration with two iSIM-90 dual channel cameras inclined around the roll axis is deemed to be the best solution, increasing the Swath Width, the footprint and enhancing the performances of the mission.

6.2 MA - GNC

6.2.1 Nominal Orbit Design

HORUS nominal orbit has been carefully selected to meet all mission requirements. The semi-major axis measures 6640.8 km , computed imposing a repeating ground track and bounding the orbit to be within the VLEO altitude range. At those very low altitudes, the air drag plays a fundamental role. Under the hypothesis of continuous thrusting and exponential atmospheric density model, the cost for drag compensation is computed for one orbit according to the eccentricity. The results show a monotonic increasing trend, indeed, as the eccentricity increases the cost increases as well. Therefore, to limit station keeping propellant, the optimal orbit shall be circular. Once the semi-major axis (a) and the eccentricity (e) have been fixed, further studies on the target orbit concern the requirement to downlink within one hour from acquisition. Ideally, a properly inclined orbit that passes over the target and the ground station is always achievable. However, at those low altitudes, the J2 perturbation becomes a driver by changing linearly the orbit Right Ascension of the Ascending Node (RAAN) with the cosine of the inclination. Therefore, a preliminary limit was imposed on the candidate solution: only orbits within range 80° - 100° inclination were considered. However, also for these orbits, a RAAN-drift compensation resulted very demanding (e.g. 40.7 m/s per day for 88° orbit). For this reason and after several iterations also considering ground station availability and downlink capacity, a polar orbit with 90° inclination was selected as the op-

timal solution. Summing up, the nominal trajectory is a polar and circular orbit at 262.75 km altitude, whose RAAN and Argument Of Perigee (ω) are selected according to target location.

6.2.2 Coverage Analysis

Up to now, only one single drone has been considered. To enlarge the overall footprint on ground, while exploiting the same orbital plane to minimize launch costs, a possible solution is to adopt a formation flight of multiple drones aligned within the same orbital plane. Starting from the selected PL, with a FoV of 2.98° , the Swath Width (SW) for a single drone nadir pointing equipped with two optical payload becomes 13.21 km , with 0.87 m of Ground Sampling Distance (GSD). By lining up several drones and putting together all different strips coming from each drone, the system is capable of reconstructing a bigger image. A graphical representation is reported in Fig. 3.

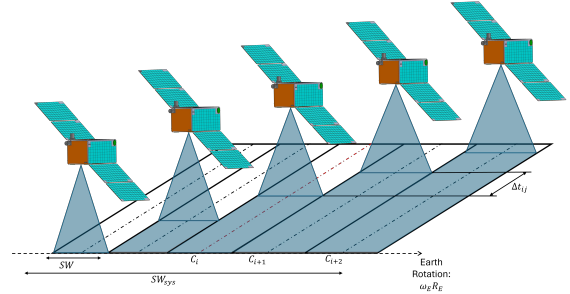


Figure 3: Nadir pointing formation flight concept

Along the design iteration process, a limit on the along-track image acquisition came out from TTMTTC sizing. A maximum of 20 consecutive images can be captured to ensure that the high level objective of downloading images within $1h$ from acquisition is fulfilled. Indeed, considering the system datarate, with more than 20 images the time required to download all the data becomes excessive. After these consideration, the coverable area function of the number of drones can be obtained. Another method to increase the coverage area, albeit reducing the spatial resolution, is to impose that the imaging mode is performed with a roll angle of maximum 28.44° . This configuration effectively increases the single SW of each drone, allowing for wider coverage areas. The coverage area function of the number of drones for the two configuration is illustrated in Fig. 4 .

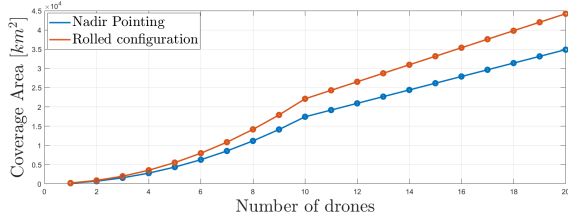


Figure 4: Coverable Area trend with increasing number of drones in nadir and rolled configuration

The drawback for the rolled configuration is that the swath width is increased renouncing to some spatial resolution, which now is set to $GSD = 0.99 \text{ m}$, still satisfying the resolution needs.

Now it is important to check the shift in true anomaly between the different drones of a formation, regardless of the roll angle assumed during imaging mode. This information is computed converting the shift on longitude that two drones have to maintain into a time delay, which, by a simple proportion, is related to the orbital angular separation of the two drones. Clearly, since the angular velocity of the Earth is faster at the equator and slower at the poles, the latitude of the target plays a crucial role. For a 20 drones formation, the angular separation between two consecutive drones is 1.9° and 10.95° for a target at the equator and at the pole respectively. In conclusion, it is possible to scale the architecture by adding drones to a formation ensuring flexibility in terms of target size. The maximum coverage area for this configuration is about 45000 km^2 .

6.2.3 Revisit Time

The selected circular, polar orbit ensures a repeating ground track. However, each drone in the formation passes over the target area only once per day. To improve revisit time a first simple solution is to put a second drone that takes an image $12h$ later. Also for this solution, the target latitude is fundamental. The revisit time can be further improved considering additional formations flying in different orbital planes, shifted in terms of RAAN depending on the time schedule requested by the customer. This new degree of freedom has to be carefully managed: indeed, some requested time of imaging and large coverage areas can lead to an overcrowding of the orbit and make the request unfeasible. Moreover, since all orbits are polar, also poles collision shall be avoided. This can be done by simply shifting the passage of a drone by some minutes.

Future developments may include giving a small eccentricity to the orbits in order to accept the intersection of two formation flights at the poles, but precisely controlling apogee and perigee so as to avoid collisions.

6.2.4 Station Keeping

To ensure correct positioning of the various satellites for optimal imaging performance, a robust station keeping strategy is implemented. The mission analysis performed so far has considered the orbital perturbation due to atmospheric drag in each mission phase. The magnitude of this effect highly depends on the Ballistic Coefficient (BC). For preliminary design two worst case scenarios for the BC are evaluated. Indeed, for initial descent from LEO to the target orbit and for disposal, low BC are preferred, while, for nominal mission operations, a high BC is beneficial. Therefore, the maximum and minimum BC are computed according to preliminary mass budget, power consumption, and cross section. Those values are employed in computing overall mission lifetime and propellant cost. Once BC computation has been performed, the air density variation, due to changes in solar flux is evaluated. This analysis is conducted using both the *Jacchia-Bowmann* and the *nrlmsise-00* atmospheric models. The worst case scenario, for minimum BC and maximum solar flux, results in a daily cost of 0.926 m/s of ΔV . For this purpose, Directional Adaptive Guidance (DAG) strategy [25],[26] has been selected to perform the orbital maneuvers. This strategy relies on the concept of virtual slot, a box of finite dimension surrounding the nominal trajectory within whom the satellite is allowed to remain. As soon as the satellite approaches the borders of this virtual slot, thrusters are activated to correct its trajectory and maintain it within the defined region. The dimensions of the slot are computed considering imaging error constraints, translating satellite non-nominal position into an error on ground. As explained in Section 6.2.2, this strongly depends on target latitude. It is worth noting that J2 and drag perturbations have no out-of-plane contributions in a polar and circular orbits, therefore defining an across-track dimension is not strictly mandatory. However, introducing this dimension could be useful for monitoring potential anomalies. In general, the worst case for slot size is for a target on the equator, in this case the virtual slot assumes the smallest dimension possible, leading to an along track dimension of 15.96 km and to an Nadir-Zenith dimension of 1.69 km .

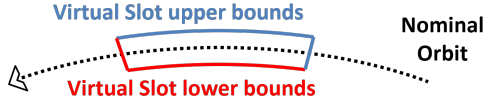


Figure 5: Virtual Slot representation

The DAG strategy works by computing the desired thrusting vector according to a weighted sum. The weights are assigned to each orbital element depending on its relevance to the mission objectives. For example, since the selected orbit for the drones is polar and the out-of-plane disturbances are minimal, the developed algorithm prioritizes corrections to the semi-major axis and eccentricity by assigning them higher weights. Eventually, the original state-of-the-art algorithm is slightly modified to set the thrusting frequency as once per orbit. Indeed, smaller but more frequent thrusting periods are required for three reasons. First, in order to ensure precise imaging, the position error shall be minimized by increasing the thrusting frequency. Second, still for accuracy purposes, thrusting during imaging is forbidden. Finally, due to thrusters high power consumption, short thrusting periods are preferred with respect to longer ones. As a result, the virtual slot is tailored in the Nadir-Zenith dimension to achieve these objectives.

To ensure algorithm robustness, a 4-month propagation is performed assessing its long-term validity. An example of a 3-day period for the semi-major axis evolution is reported in Fig. 6.

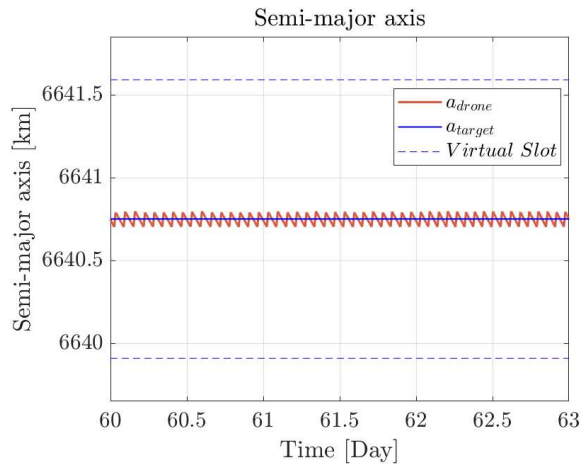


Figure 6: Semi-major axis evolution from day 60 to day 63

6.3 TTMTTC

One of the most critical subsystems for the HORUS mission is the TTMTTC, as it plays a key role in achieving a primary high-level objective: downlinking images within one hour from their acquisition.

The HORUS TTMTTC architecture consists of a classical system featuring a transmitter, a receiver, and an amplifier connected to four patch antennas, ensuring redundancy and reliability. Given the mission requirements for high data rates, S-band frequency was selected due to its optimal trade-off between transmission power and atmospheric losses at low orbital altitudes.

6.3.1 Communication Strategy

The initial phase of the subsystem design involved the selection of the communication strategy and the definition of the ground segment architecture. For HORUS, a direct space-to-ground link was chosen as communication strategy, motivated by the stringent requirements of global coverage and rapid data transmission. Operating at very low Earth orbit altitudes results in high relative velocities with respect to ground stations, imposing significant limitations on antenna selection for ground-based tracking. Consequently, only antennas capable of rapid tracking can be utilized effectively, and large-diameter antennas, which typically offer limited agility, were excluded. Therefore, an upper diameter limit of 8 meters for antennas was established.

To determine the optimal ground stations for the HORUS network, a comprehensive coverage analysis was conducted. The goal was to ensure that, for any randomly selected target location on Earth, the downlink of the payload data could be guaranteed within one hour from image acquisition, meeting the mission's strict latency requirements. This analysis also accounted for additional constraints related to contact windows, data budget, and TTMTTC architecture. Considering the lowest data budget scenario and the maximum imaging load (40 images, 20 for each camera), HORUS requires at least 53 seconds for full data downlink. Additionally, accounting for 30 seconds needed to initiate ground station contact and another 30 seconds for spacecraft release results in a minimum required visibility window of approximately 2 minutes. Initially, the contact analysis relied exclusively on ESA's Estrack ground station network. However, this configuration failed to satisfy the mission requirements, as the spacecraft could not reliably achieve complete data downlink within one hour post-acquisition, as visible in Figure 7.

Indeed, relying only on the Estrack network would necessitate relaxing the latency requirement to approximately 2 hours and 25 minutes, an unacceptable compromise for the mission objectives.

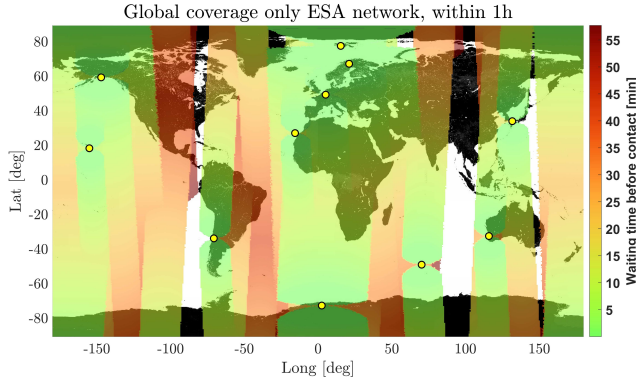


Figure 7: Estrack global coverage within 1h

Consequently, to enhance coverage and reduce latency, ground stations operated by private companies were integrated into the network. European companies SSC[©] and LeafSpace[©] were selected as additional ground segment providers. Figure 8 illustrates the results of this enhanced network analysis, showing ESA's ground stations marked in yellow and commercial ground stations indicated in cyan.

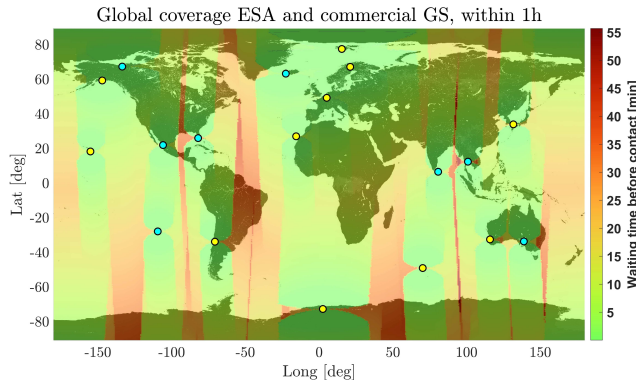


Figure 8: Global coverage ESA and commercial network within 1h

6.3.2 Contact Analysis

The analyses confirm that a single-drone configuration is feasible for the mission. However, complications arise when multiple spacecraft operate simultaneously. In particular, scenarios with two drones flying the same orbit but with a defined phase difference present significant challenges, especially when imaging equatorial

targets. To produce coherent surface images from combined data, the drones must be closely spaced along the orbit. This leads to very short time intervals, in the order of tens of seconds, between their respective passes over the ground station. Such short intervals create downlinking difficulties, as the ground station is still busy receiving data from the first drone when the second one arrives.

Detailed analyses indicate that while any global location can reliably be serviced by at least one drone, optimal regions, such as the Middle East, can simultaneously support up to 19 drones with careful coordination. A detailed example illustrating these considerations is provided by the case of a 5-drone formation targeting the Paris Metropolitan Area (approximately 4,200 km²). In this scenario, drones are strategically assigned to different ground stations, allowing each spacecraft to maintain the required one-hour latency for data downlink. The visibility windows and corresponding contact durations for each drone and ground station are illustrated in Figure 9. For instance, Drone 1 connects immediately after imaging to the Redu ground station for around 170 seconds, while Drone 2 makes contact with the Troll Antarctic ground station approximately 120 seconds before the closure of the one-hour window.

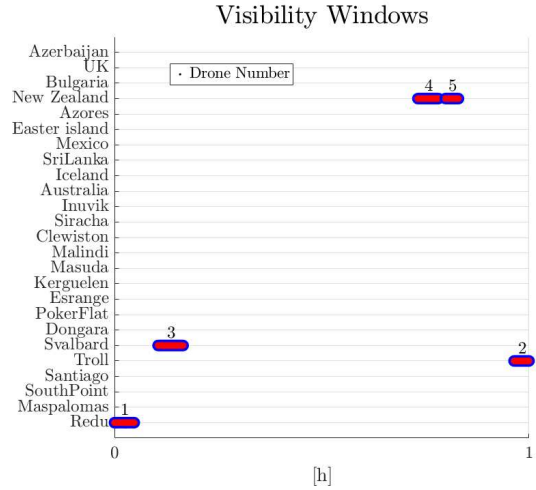


Figure 9: Visibility study over Paris

Conversely, challenges arise when larger drone formations target broader areas, such as the Eastern Mediterranean Sea (approximately 35,000 km²). In one studied scenario involving 20 drones imaging the region around Cyprus, 19 drones successfully achieved timely ground station contacts within the one-hour limit, while one drone could not, resulting in an incomplete coverage area as shown in Figure 10. The remaining images

were downlinked within two hours of acquisition. Addressing such gaps requires careful mission planning, potentially repositioning individual drones into alternative orbits to ensure seamless coverage.

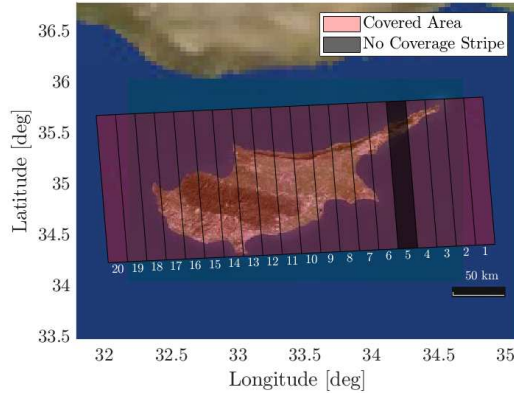


Figure 10: Visibility study over Cyprus

6.4 Other subsystems

6.4.1 Propulsion Subsystem

The Propulsion Subsystem (PS) is designed to face the two main propulsion needs of the HORUS spacecraft: the orbit insertion manoeuvre and the station keeping manoeuvre.

The orbit insertion manoeuvre is the most critical period for the propulsion subsystem, since the satellite needs to position itself from the release orbit to the correct orbit, requiring approximately a ΔV of 137.1 m/s . The station keeping manoeuvre is instead mainly needed due to the air drag present in the VLEO orbit, which costs 0.81 m/s per day. The total ΔV of the mission is low, as it can be measured in hundreds of m/s . Moreover, the thrust direction is only along one axis, therefore the choice relies on a single primary propulsion system. The propulsion system chosen is the SPS-25 Xenon-powered Hall Effect Electric Thruster (HEET), developed by the Space Electric Thruster Systems (SETS) company [27].

The main advantages of this propulsive system are the limited mass of propellant required, essential for VLEO missions, and the manoeuvre duration. Moreover, the relatively low power consumption, the wide thrust range, and the high specific impulse offered by this propulsion system confirm the choice. The propulsion subsystem is entirely commercially bought from SETS, apart from the tanks, which are in-house developed. This choice was made after careful consider-

ations, leading to the intention of building the tanks in order to minimize the volume while maximizing the tank pressure and the propellant mass, which is approximately 8 kg . The drawback of choosing this kind of propulsion subsystem is the degradation of the thruster. In fact, the thruster lifetime is the most limiting factor of the spacecraft lifetime, leading to a maximum operational period of approximately 1.5 years.

6.4.2 EPS

The Electrical Power Subsystem (EPS) of the HORUS mission is designed to ensure reliable power generation, storage, and distribution in a VLEO environment. The system is sized according to mission requirements, operational mode power requests, and orbital constraints. The power budget is margined at both component and system levels, following ECSS standards. The baseline power architecture consists of solar panels as the primary energy source, complemented by a secondary battery system to provide energy during eclipses and peak demand periods. A set of six panels, two of which are body mounted, is the output of the sizing process. Triple-junction solar cells with high BoL efficiency are selected. To optimize energy extraction, Maximum Power Point Tracking (MPPT) is implemented. A "Planar" Sun-pointing algorithm is also implemented, which maximizes power generation by performing slew maneuvers solely about the roll axis. This approach minimizes the spacecraft's cross-sectional area while reducing the Sun-solar panel incidence angle, thereby enhancing power production. Instead, the battery employs Li-ion cells, a well-established technology in the space field. Extra solar cell strings and battery redundancy ensure continued operation in the event of partial failures.

6.4.3 ADCS

The ADCS subsystem is responsible for satellite correct attitude determination and control. The design of the subsystem is mostly constrained by the high pointing accuracy required during the imaging mode. To ensure its 0.1° requirement and redundancy constraints, the final subsystem architecture includes: two star sensor, five fine Sun sensors, two gyroscopes, two magnetometers, a GNSS receiver, four RW in pyramidal configuration, and a three-axis magnetorquer. Since the satellite is at a very low altitude, the subsystem guidance is designed aiming to minimize HORUS cross section. This task is performed exploiting mostly rotations about the roll axis, keeping pitch and yaw angle as close to zero

as possible in most of the modes. Regarding the navigation solution, during the imaging mode, the attitude is directly retrieved via star sensor, while in almost all the remaining modes a quaternion-based Multiplicative Extended Kalman Filter is employed. This choice ensures to have a very accurate estimate also without star sensor measurement, enhancing SA array pointing for power production and antenna pointing for telecom. The control solution involves a well established Linear Quadratic Regulator (LQR) algorithm for telecom and Sun pointing purposes, an augmented B-dot for detumbling, and a Linear Quadratic Integral (LQI) algorithm for the high performance imaging mode.

6.4.4 TCS

The Thermal Control Subsystem is designed to ensure that all spacecraft components remain within their operational temperature ranges using a quasi-passive thermal management system. Given the varying heat fluxes encountered across different mission orbits, a highly flexible thermal design is essential. Radiators are integrated into the bottom surfaces of both the solar arrays and the main body. To minimize exposure to external sources, only the solar assemblies and radiators interact with the environment, while the core body remains thermally isolated through insulating spacers. Heat switches, operating on a temperature-activated logic, regulate cooling based on internal dissipation and external conditions. Thermal regulation is further optimized through the overall attitude control strategy, ensuring alignment with power production requirements. Additionally, thermal straps connect solar panels to radiative surfaces, optimizing power generation efficiency. Coatings and surface treatments are applied considering degradation effects in the VLEO environment. The implementation of a glass cover to enclose the spacecraft aperture is being studied to minimize temperature fluctuations in the PL caused by the opening inside the spacecraft body, avoiding undesired thermoelastic effects affecting the performance during the imaging.

6.4.5 OBDH

The On-Board Data Handling subsystem is responsible for managing, processing, and storing all data from both the platform and the payload. The selected architecture is a federated bus configuration, consisting of a main OBC that runs the flight software and manages the overall platform control, and a secondary OBC dedicated specifically to the AOCS subsystem, running the GNC algorithms. The CAN bus has been chosen

as the primary bus due to its high reliability and the interfaces available on the OBCs, while a SpaceWire has been selected as a high-speed bus to minimize latency between payload data acquisition and downlink. Redundancy is ensured both on the hardware and on the software side. One of the main criticalities of this subsystem is that it needs to be capable to withstand the VLEO radiation environment for the whole mission duration. For this reason a Total Ionization Dose analysis has been conducted and a 1mm aluminum shielding will be produced to mitigate the risk of OBC failures.

6.4.6 Structure

The structural section of the HORUS spacecraft is designed to withstand the extreme VLEO environment. The structure can be divided into a primary and a secondary structure. The primary structure is composed by the main body, a $330 \times 330 \times 830\text{mm}$ cuboid, entirely made by the Aluminum alloy Al7075-T6, featuring a 1.6mm thickness. The choice of the material was conducted considering the structural robustness of the spacecraft and the environmental degradation caused by the VLEO environment. The secondary structure is mainly composed of the two deployable solar arrays, each one made by two solar panels. The solar panel has a length of 830mm and a width of 370mm. The role of the two solar arrays is to maximize the power generation for the spacecraft and dissipate the excessive heat due to the solar exposure. In order to ensure a reliable and robust platform for the spacecraft, an exhaustive set of structural analyses is conducted. The study consisted of modal analyses, quasi-static analyses, sine and random vibration analyses, and linear buckling analyses, based on the Vega-C and Ariane 6 user manuals. Eventually, additional analyses on the panels were performed, as these are a sensitive component of the spacecraft.

6.4.7 Configuration

The configuration of the HORUS spacecraft is studied to achieve optimal performance in VLEO, ensuring that all subsystems operate as required. The external configuration was designed to meet the structure, power generation, and heat dissipation requirements, with also a deep study on sensors and antennas positioning. The main body is connected with the solar arrays at the two lateral edges, with the Sun-pointing faces fully covered by solar cells to ensure the most efficient power generation. The opposite side is instead covered with radiators, in order to achieve the best heat

exchange possible. A cross section study was conducted in order to ensure the best spacecraft performances. The square-shaped cross section is chosen to simplify the structure, minimizing the most the area in order to decrease the aerodynamic drag. After careful analyses, no pyramidal shape is evaluated, as it has been assessed that no advantage can be gained from that solution. Eventually, a meticulous analysis was conducted for the positioning of the sensors and the antennas. In fact, three patch antennas are placed in the lower part of the spacecraft to guarantee a correct link with the ground and maximise the visibility windows. Moreover, a fourth antenna is placed in the upper part of the spacecraft to ensure the connection also in the first phases. The two star sensors are perpendicularly placed on the upper surface and on the front surface, to operate under the best possible conditions. Additionally, five Sun sensors are positioned on the main faces of the spacecraft for ADCS purposes. Eventually, a GNSS antenna is placed on the upper part of a solar array. The internal configuration main objectives are to minimize the CoM shift meanwhile the tanks empty, and to achieve optimal heat fluxes between components, in order to satisfy the requests of both ADCS and TCS. Eventually, a deep analysis on the payload position was conducted. Indeed, being it a high resolution camera, it was mandatory to minimize as possible the disturbances, therefore the payload is strategically positioned as close as possible to the CoM.

7 Cost

The HORUS mission has been developed with cost-efficiency as main design principle, pointing to deliver high-performance Earth observation capabilities at a sustainable price. The total cost is organized into three main components: the space segment, the launch phase, and operations. For the space segment, the design adopts a compact and modular satellite, with costs estimated through a parametric approach based on Cost Estimating Relationships, by employing the Small Satellites Cost Model [28]. The use of European COTS components and high-TRL technologies helps reduce development costs, while in-house development is limited and reserved for critical subsystems, specifically the propulsion system tanks, the ADCS software and the TCS heat switchers. Fig.(12) represents the cost distribution of the space segment.

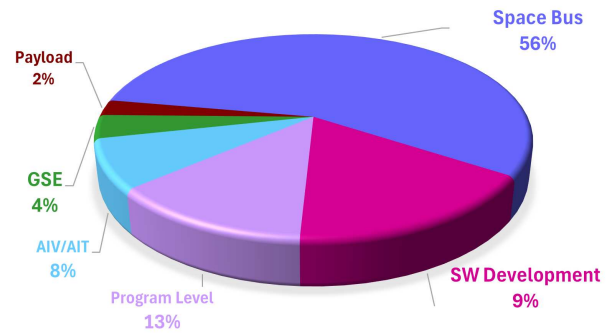


Figure 12: Cost distribution: first drone case

Launch expenses are minimized by selecting small-lift European launchers and leveraging rideshare opportunities. Operational costs are estimated for a year and a half nominal mission duration and include ground infrastructure, personnel, and data processing. A 20% margin is applied across all segments to account for uncertainties in material availability, supplier timelines, and logistics. The total estimated cost with margin for a single HORUS unit, including space segment, launch, and operations, is approximately 26.47 M€. In the case of a ten-unit deployment the aggregated cost reaches 99.49 M€. These highlight the mission's scalability and the cost benefits of series production. The modular architecture and the ability to deploy only the required number of units based on mission needs ensure a flexible and economically sustainable solution for responsive Earth observation services.

8 Possible applications

Given its advanced capabilities, HORUS can address a diverse range of Earth Observation applications. First, its ultra-high spatial resolution combined with rapid revisit times enables precise environmental monitoring, making it ideally suited for tracking deforestation, assessing crop health, monitoring biomass loss, and other climate change-related tasks. Second, the capability to deliver imagery within one hour from acquisition is a game-changer in real-time disaster response, effectively supporting search and rescue missions, and enabling accurate, timely assessments of flood impacts or wildfire perimeters. This capability will be further enhanced as launch opportunities from European territory increase. Moreover, the use of airborne launch platforms, once commercially available, could reduce the time to fly from weeks to just few days, significantly improving responsiveness to emerging customer needs and enabling new time-sensitive applications. HORUS also offers

strong potential for dual-use operations, supported by the fact that it only employs European technologies. It is well-suited for maritime surveillance, such as monitoring maritime traffic and detecting illegal activities, as well as ensuring border and infrastructure security, all without the cost and complexity of large permanent satellite constellations. Ultimately, the long-term mission of HORUS is to make this cutting-edge technology accessible not only to institutional users but also to non-governmental organizations and commercial entities, opening the door to customized campaigns across scientific, humanitarian, and commercial domains.

9 Conclusions and Future Developments

The HORUS mission study has demonstrated the technical and economic feasibility of deploying high-resolution Earth observation satellites in VLEO. Detailed analyses and studies were conducted, leading to a coherent and integrated satellite design. The study included orbital dynamics, mission operations, ground segment planning, and cost modelling. By combining innovative solutions with flight proven components, HORUS offers a suitable and innovative platform ready for real-world implementation. The mission proved to be a valuable testbed for advancing the TRL of critical components and validating their long-term performance in the VLEO environment. In particular, the components already tested in LEO will be re-evaluated in VLEO, to provide strategic insight into

material durability, system efficiency, and environment resilience. Currently, several future developments are being considered to enhance the mission's capabilities. A promising future approach is the use ABEP, which can extend the mission duration by collecting and using residual atmospheric gases as propellant, allowing a reduction of mass at launch and increasing the overall mission lifetime. Further developments are expected in communication capabilities, by adding inter-satellite links and relay satellite constellations, which will enhance downlink data rate.

From a design point of view, more advanced studies will be explored: aerodynamic shape optimization to reduce drag, the use of aerodynamic control surfaces for passive or hybrid attitude control, and the application of innovative materials or protective coatings to mitigate surface degradation from ATOX and UV exposure. All of these elements will contribute to improving the stability, efficiency, and durability in VLEO conditions.

Finally, future efforts will focus on the development and adaptation of mission facilities, which will evolve to support the rapid deployment and integration of new units, primarily through the use of parallel production. This will include upgrading the ground segment to increase data volumes and contact frequency. These advancements will allow HORUS to evolve from a single demonstrator into a fully deployable, scalable, and reusable Earth observation system, establishing the beginning of a new generation of mission in the VLEO environment.

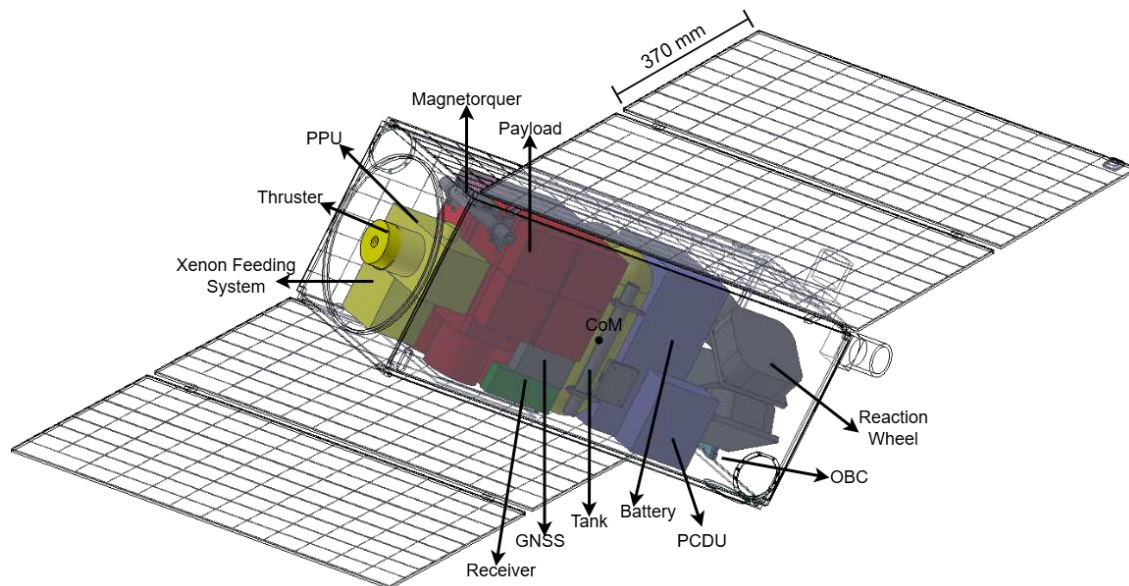


Figure 11: HORUS internal configuration

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